

AI-GIS Integrated Vulnerable Population Monitoring and Aid Recommendation System for Welfare Services

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Project Proposal Report
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1. Declaration

I guarantee that I am the sole author of this proposal report. Unless I specifically stated otherwise, I have never used or submitted any other person's work for another degree or diploma. To the best of my knowledge, no one else has authored or published any portion of this report without their permission.

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2. Abstract

Sri Lanka's social welfare programs are currently plagued by serious problems because the majority of the work is done by hand, resources are not distributed efficiently, and information about those in need is scattered across several sites. This study proposes the development of a smart computer system that combines the mapping technology known as Geographic Information System (GIS) with Artificial Intelligence (AI). This method will assist in identifying and displaying the residences of vulnerable populations, including patients with chronic illnesses, pregnant women, and individuals with disabilities. Technology can identify places where social welfare support is most required by evaluating location-based data and forecasting who could require assistance the most.

Policymakers and bureaucrats will be able to make more informed decisions on how to distribute funds and provide more efficiently by utilizing this cutting-edge AI and GIS platform. To make it simpler to track and address social needs, the system will provide interactive maps that display "hotspots" of welfare need. In general, this strategy seeks to improve accessibility (making it simpler for individuals in need of aid), promote efficiency in Sri Lanka's welfare systems, and increase transparency (by making it evident where support is required). A versatile mapping and analytic tool that promotes forward-thinking governance will be the result, enabling Sri Lanka's social welfare system to reach a wider audience and have a greater positive influence.

Keywords: Sri Lanka social welfare, Geographic Information System (GIS), Welfare demand hotspots, welfare service mapping

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6. Introduction

Sri Lanka has a wide range of welfare programs to aid those in need, including those with disabilities, the elderly receiving pensions, and those in need of food assistance. However, these programs have issues. Numerous operations are still completed by hand, and the data is kept in various locations. Delays and errors result from this. Furthermore, resources are not often used to their full potential because there is no system that displays up-to-date information about the residences of those in need of assistance.

The goal of this project is to develop a smart computer system that makes use of Geographic Information System (GIS) mapping technology and Artificial Intelligence (AI). The system will gather and analyze information from many government sources regarding people's welfare, income, and health. It will identify vulnerable populations and forecast areas with the greatest need for assistance. The system would produce easily comprehensible maps and data to assist officials in making quicker and more informed decisions regarding relief distribution.

Three major user groups will gain from the system's operation as a component of this integrated welfare platform:

Citizens: People will get better access to welfare programs with more accurate information, speedier approvals, and location-specific help.

Welfare Officers: They can use the system to easily manage data, quickly identify areas that need more attention, and keep track of who needs help. This will make their duty easier and more accurate.

Policymakers: To better manage budgets, identify areas that require additional support, and assess the effectiveness of initiatives, they will receive comprehensive reports and visual maps. The entire welfare system will become more transparent and efficient as a result.

All things considered, this AI and GIS system would improve Sri Lanka's welfare services by offering real-time location data and forecasts. It will reduce errors and delays, increase system transparency, and ensure that the right individuals get aid when they need it. This will enhance the country's social safety net in the future and encourage equitable prosperity.

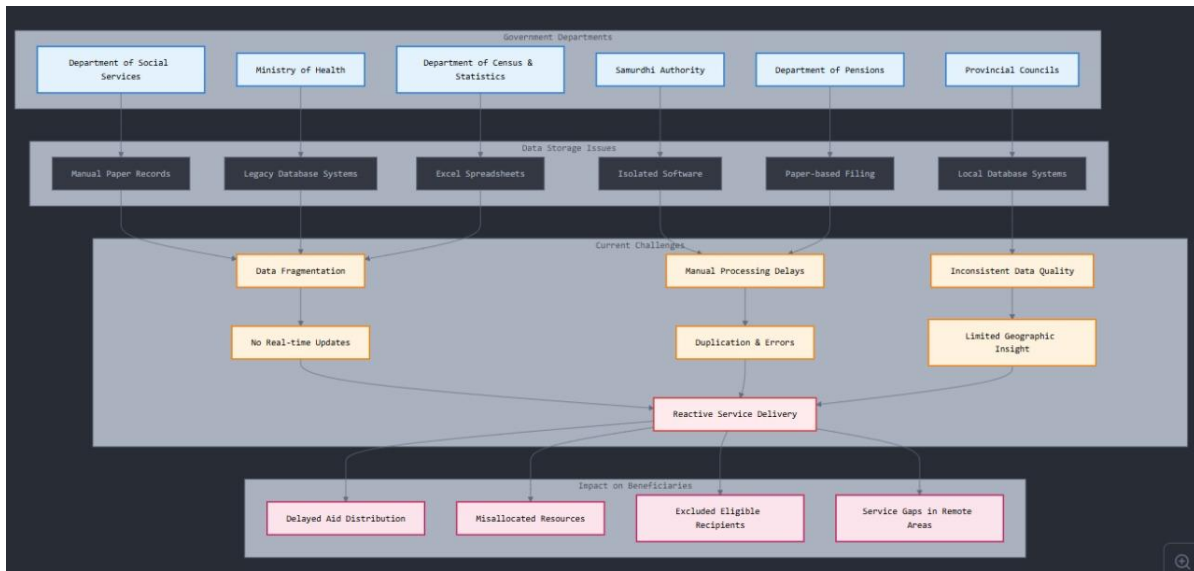


Figure 1 = Current Sri Lanka Welfare System Data Flow and Challenges

The severe inefficiencies in Sri Lanka's disjointed social welfare data management system are shown in Figure 1, underscoring the pressing need for an AI-GIS integrated solution. Social Services, Ministry of Health, Census & Statistics, Samurdhi Authority, Department of Pensions, and Provincial Councils are the six main government departments that currently handle welfare information. With antiquated techniques like handwritten paper records, legacy databases, Excel spreadsheets, and locally managed systems, each department functions independently, posing serious operational difficulties that seriously reduce the efficacy of social service delivery.

Three main systemic issues are caused by the dispersed data storage: irregular data quality results in incorrect beneficiary identification, manual processing delays services for weeks, and data fragmentation hinders thorough beneficiary profiling. The lack of real-time updates, frequent data duplication and errors, a lack of spatial knowledge into welfare distribution, and primarily reactive service approaches that don't foresee community requirements are some of the operational limits that are exacerbated by these problems.

These systemic problems have detrimental effects on vulnerable populations that depend on social services. Incomplete information leads to the misallocation of resources, eligible recipients are denied access to the program due to verification problems, aid distribution for beneficiaries is delayed during critical periods, and service gaps persist in remote areas where manual data collection is most challenging.

This picture makes it abundantly evident why changing Sri Lanka's welfare administration from a disjointed, reactive strategy to an integrated, proactive delivery model requires a centralized, AI-powered, spatially aware system.

7. Background & Literature Review

Vulnerable populations, such as low-income families, individuals with impairments, the elderly receiving pensions, and those in need of nutritional support, are the focus of Sri Lanka's social welfare system. A number of significant programs are run by the government, including food assistance programs, disability payments, pensions for the elderly, and Samurdhi financial aid. These programs aim to improve the living conditions and social protection of those who are less fortunate [1].

Despite these initiatives, the management of social assistance programs faces many challenges that limit their efficacy. The fact that welfare data is scattered across numerous government departments and agencies is one of the main obstacles. Each department keeps its own records, frequently in different formats and databases. It is challenging to get an accurate picture of beneficiaries because of the gaps and inconsistencies caused by this fragmentation [2]. Furthermore, a lot of procedures, including the submission, verification, and acceptance of applications, are still done by hand or with antiquated technology, which slows down service delivery and raises the possibility of errors [3].

Another significant issue is beneficiary misidentification. Some ineligible individuals can get aid while legitimately needy residents are left out due to gaps and errors in data verification. This problem is exacerbated by the lack of an integrated system for automatic eligibility validation and cross-checking [4]. Furthermore, policymakers and welfare authorities do not have access to up-to-date, location-based data. Targeted interventions and effective resource distribution are challenging without real-time information into high-need areas, resource use, and the whereabouts of vulnerable populations [5].

In order to enhance delivery service, other countries with comparable welfare issues have implemented digital transformation projects. The use of e-governance, GIS, and AI has completely changed welfare management. To improve transparency, reduce fraud, and expedite benefit distribution, India's Aadhaar-based welfare system combines biometric, financial, and demographic data [6]. In order to better focus aid, Bangladesh has used AI-based techniques to assess social and health variables spatially [7]. These solutions use Geographic Information Systems (GIS) to visualize areas that need immediate attention, together with artificial intelligence approaches like clustering algorithms to group beneficiaries and classification models to forecast eligibility [8].

Predictive analytics is a new area in this discipline. By evaluating past and present data, AI models can predict welfare demand, helping governments better manage their finances and plan ready for natural disasters like floods and pandemics [9]. Such predictive ability lowers administrative expenses and delays while guaranteeing aid to those most in need.

However, there has been little uptake of AI-GIS integrated solutions in Sri Lanka's welfare environment. Current systems frequently lack advanced analytics capabilities and are siloed. Despite the collection of important demographic and socioeconomic data, it is not fully leveraged to produce spatial or predictive insights [10]. As a result, welfare programs experience underutilization of resources, ineffective targeting, and delayed responses.

The lack of a centralized, AI-powered GIS platform is a significant flaw in Sri Lanka's welfare management system. By combining census, health, and beneficiary data, spatial clustering could be used to identify vulnerable groups, such as pregnant women, individuals with chronic illnesses, and people with disabilities. Future welfare demand and new hotspots could be identified using predictive models [9].

By improving the accuracy of beneficiary identification, increasing the transparency of resource allocation, and enabling real-time policy decisions, the deployment of such a platform has the potential to completely transform welfare government in Sri Lanka [6][8]. Welfare officials could quickly spot irregularities with interactive maps and dashboards, and policymakers could use data-driven insights to optimize welfare spending and coverage [7].

To sum up, integrating AI and GIS technology into Sri Lanka's welfare administration can help to improve efficiency, lessen exclusion and benefit leakage, and promote fair socioeconomic development. This project attempts to create a more intelligent, inclusive, and responsive system to safeguard vulnerable populations by developing an innovative AI-powered GIS platform that is suited to Sri Lanka's welfare concerns [1][10].

8. Research Gap

Sri Lanka offers a number of social welfare programs to help the elderly, disabled people, and low-income families. However, the absence of a single digital platform that collects data from multiple government agencies and social programs presents major challenges for these projects. Data is now scattered and often processed manually or across multiple systems, which can lead to errors, delays, and erroneous information. It is difficult to swiftly and precisely identify who needs help and where because of this fragmentation.

Another major gap is the absence of real-time, geospatially informed data visualization tools that can map vulnerable populations and show welfare demand across multiple districts or areas. Without these geographic insights, policymakers cannot quickly identify high-need areas or address new welfare issues. The lack of up-to-date, location-based data undermines the efficiency and targeting of social welfare services.

Furthermore, Sri Lanka's welfare management system does not currently use predictive models or artificial intelligence (AI) to forecast future benefit demand. Predictive analytics can help anticipate future aid needs by using historical data and demographic trends, allowing for more proactive resource distribution. Because the system is now largely reactive, there may be insufficient resources or excessive spending in the wrong areas.

Lastly, these issues raise the possibility of financial mismanagement or misuse, delays in benefit delivery, and inaccurate policy judgments. Integrated AI-GIS solutions that offer thorough, real-time, and predictive welfare management are still lacking in Sri Lanka, despite the country's initial digitalization efforts. By creating a centralized AI-powered GIS platform that can estimate welfare demands, integrate data sources, and facilitate geographic visualization, this project will close these gaps and enhance social welfare governance's efficiency, justice, and transparency.

Table 1: Comparison between existing systems

Feature/Aspect	Current Sri Lankan System	Proposed AI-GIS System	International Examples (India/Bangladesh)
Data Integration	Fragmented across multiple departments	Centralized spatial database	Unified digital platforms with biometric integration
Processing Method	Manual paperwork and data entry	Automated AI-powered processing	Digital workflows with automated verification

Geographic Visualization	No spatial mapping capabilities	Real-time GIS maps and heatmaps	Geographic targeting with spatial analytics
Beneficiary Identification	Manual verification, prone to errors	AI classification algorithms	ML-based eligibility determination

6. Research Problem

An intelligent computer system that integrates Geographic Information System (GIS) with Artificial Intelligence (AI) is required in Sri Lanka. This method should assist in identifying the residences of those who are at risk, forecasting the type of social assistance that they will require, and assisting the government in making better decisions. Making social welfare programs more effective, transparent, and equitable for all those in need is the aim.

One major challenge is the fact that data about people's health, wealth, and well-being originates from multiple sources. These data sets are stored in multiple formats and locations, making their combination difficult. The new system must integrate these diverse data types into a single, reliable, well-structured location-based database that can be updated frequently.

The development of AI systems that can evaluate this aggregated data to determine who needs assistance and where is another significant issue. In order for the government to prepare ahead of time and allocate resources before issues worsen, these AI models should also forecast future welfare demands. This capacity for prediction will assist in the transition from responding to issues to averting them.

Last but not least, the system has to provide clear maps and reports that policymakers and government representatives may utilize right away to improve their decision-making. At the same time, it is crucial to adhere to all national data protection rules and safeguard any personal information. The platform should employ robust security measures to safeguard beneficiary privacy and only permit authorized individuals to access sensitive data.

1. In what ways might artificial intelligence be applied to precisely forecast welfare program eligibility?

Due to the current manual or rule-based eligibility evaluations' high error rate, some eligible citizens are denied benefits while ineligible ones are granted them. Creating AI models that can evaluate intricate socioeconomic and demographic data in order to guarantee equitable and precise eligibility forecasts is a problem. Finding suitable machine learning methods and allowing the system to adjust to various citizen profiles are necessary for this.

2. How may welfare recipients receive multilingual, real-time assistance?

Due to inadequate internet literacy and linguistic challenges, many welfare applicants have trouble understanding application procedures or checking their status. The difficulty lies in creating a multilingual chatbot driven by AI that can converse fluently in English, Tamil, and Sinhala. A system like this can increase inclusivity, close communication gaps, and facilitate the utilization of social services.

3. How can fraud detection and data visualization equip policymakers with the tools they need to make decisions quickly?

There are currently no efficient tools available to policymakers to track program utilization, identify fraudulent activity, or instantly map risk areas. Creating AI-powered dashboards that combine predictive analytics and geographic information systems (GIS) is a problem. These dashboards ought to offer useful information, improve openness, reduce resource waste, and facilitate the most efficient distribution of welfare benefits.

4. How can welfare services incorporate healthcare recommendations?

In addition to cash help, vulnerable beneficiaries frequently need healthcare aid; yet, current welfare systems do not offer individualized medical advice. The difficulty lies in creating a recommendation system that links citizens with pertinent healthcare providers while taking into account their unique health profiles. Welfare initiatives will be more effective overall and have a greater social impact if healthcare advice are incorporated.

Sri Lanka's systems are still antiquated and disjointed, in contrast to nations like Bangladesh and India that have effectively incorporated e-governance, Geographic Information Systems (GIS), and Artificial Intelligence (AI) to simplify welfare services [5][6]. As a result, there is now a considerable lag in the adoption of contemporary digital systems that can detect fraud, do predictive analysis, and provide real-time monitoring.

10. Main and Sub Objectives

10.1 Main Objectives

Modern technology can be used to more accurately identify those in need of assistance and make sure that aid is distributed to them in a timely and equitable manner, thus improving social welfare services in Sri Lanka. The goal of this project is to develop a smart computer system that integrates geographic information system (GIS) mapping with artificial intelligence (AI). People with impairments, the elderly, expectant mothers, and people with chronic illnesses are among the vulnerable categories that this system will assist in identifying and monitoring. The platform seeks to improve the speed, equity, and efficiency of social welfare delivery nationwide by mapping these groups and forecasting their welfare requirements.

- Create and implement a GIS system with AI integration that can locate and map Sri Lanka's most vulnerable populations.
- Gather and combine demographic, health, and socioeconomic data from many government sources to create a single spatial database.
- Create machine learning algorithms to identify disadvantaged populations and forecast areas with high welfare needs in the future.
- Create user-friendly GIS dashboards and visualizations to assist welfare officials and legislators in making informed judgments.
- Make the system dependable and easy to use for all users while maintaining data security and privacy compliance.

10.1 Sub Objectives

A number of crucial actions will be taken to gather, examine, display, and safeguard the welfare data in order to accomplish the primary objective. These actions will guarantee that the system functions efficiently and satisfies the requirements of every user.

- **Gather and link data from various sources.**

Compile comprehensive data on people's living arrangements, health, and social and economic standing. Various government agencies and departments will provide this data. The objective is to compile all of this data into a single, easily navigable map-based database so that the system is aware of the various vulnerable categories and where they are.

- **Create intelligent computer models to identify and forecast needs.**

Analyze the combined data using AI and machine learning methods to determine the locations and identities of the vulnerable populations. Additionally, these models will forecast future welfare needs, allowing the system to identify which areas may require additional assistance in the near future. This aids in planning and preparing welfare assistance before issues worsen.

- **Create dashboards and visualizations that are simple to grasp.**

Create straightforward, interactive dashboards and maps that illustrate the locations of those who are at risk and how their needs evolve over time. Government officials and legislators will be able to swiftly comprehend the issue and make better decisions about where and with whom to provide assistance thanks to these visual aids.

- **Protect data and make the system user-friendly.**

Use robust security measures and abide by Sri Lanka's privacy regulations to safeguard sensitive personal data. Make sure the system is easy to use so that welfare office staff members without specialized technological knowledge may utilize it with ease.

11 Methodologies

11.1 Overall System

The approach describes the procedures and methods utilized to create the Geographic Information System (GIS) platform with AI integration for enhancing social welfare services in Sri Lanka. Data collecting, AI modeling, spatial analysis, visualization, and data security are all combined to produce a powerful, intuitive solution that facilitates resource allocation and decision-making.

- **Gathering and Combining Data**
Collect and purify demographic, health, and socioeconomic data from several government sources. To precisely map vulnerable groups, compile all of this data into a single, cohesive spatial database.
- **Creation of AI Models**
Identify vulnerable populations and forecast future welfare needs and hotspots using machine learning, which gets better with time and more data.
- **Dashboard Design with GIS Visualization**
Make user-friendly dashboards and maps that highlight areas in need of assistance, providing officials with real-time information to help them make better decisions.
- **Compliance, Security, and Privacy of Data**
Use robust security measures to safeguard personal information, make sure the system complies with national privacy regulations, and grant access to only authorized users.
- **Testing the System and Training Users**
Verify the system's accuracy and usability in-depth. Educate policymakers and welfare workers, then use their input to make the platform better.

Figure 2:proposed AI-GIS Integrated System Architecture



11.2 Individual System Diagram

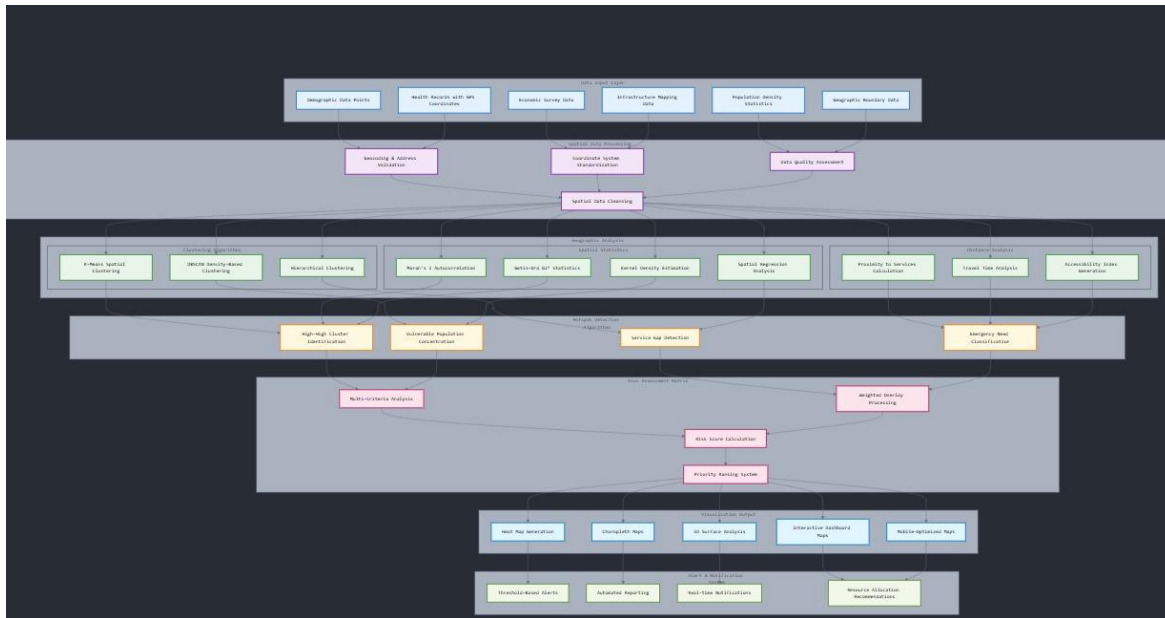


Figure 3: GIS Spatial Analysis and Hotspot Identification Process

In order to give welfare officers and citizens quick, accurate information, this component combines eligibility prediction with healthcare suggestions. The following interfaces and subcomponents make up the module I oversee:

1. Gathering and Preparing Data

combines health and socioeconomic information from several sources.
handles missing values, cleans, de-duplicates, and engineers features for machine learning models.

2. Module for Eligibility Prediction

Uses supervised classifiers (e.g., Decision Trees, Logistic Regression, Random Forests) to estimate eligibility.
generates an eligibility score or judgment that, when appropriate, includes justifiable variables.

3. Module for Healthcare Recommendations

matches beneficiaries with appropriate healthcare services using rating and retrieval strategies.

The Post-Therapy Activity Assignment subcomponent works with therapists to assign stress-relieving activities that are specific to the patient's situation during the post-therapy phase

4. Interface and Integration

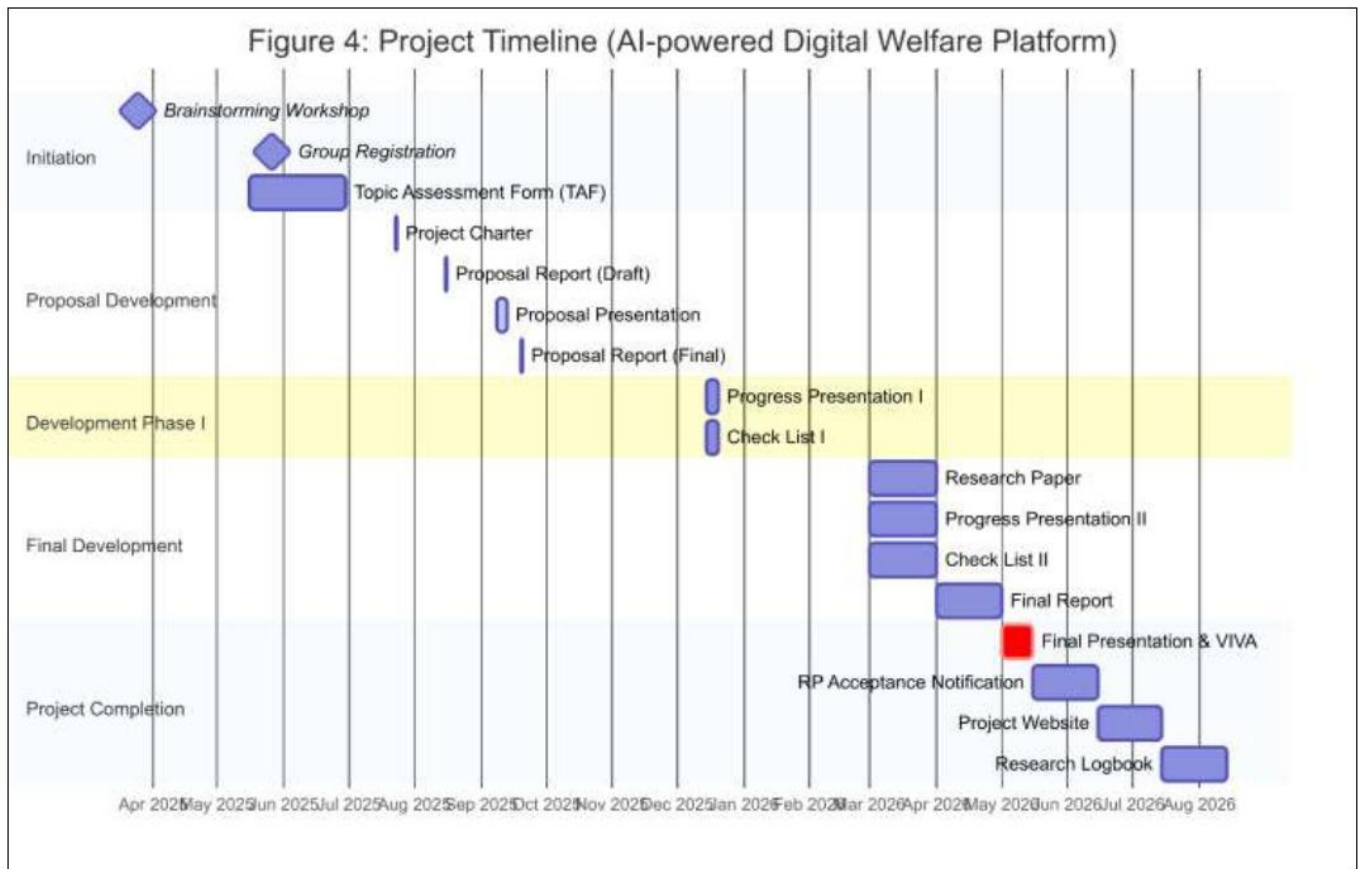
ML outputs are connected to front-end apps via the API layer.

Citizen Portal: appointments, document upload, status tracking, and a multilingual chatbot in Sinhala, Tamil, and English.

Officer Dashboard: case view, audit trail, overrides, alarms, and verification queue. interactivity and feedback for end users in real time.

12. Gantt Chart

Figure 4: Gantt Chart



13. Project Requirements

13.1 Functional Requirements

- **Integration of Spatial Data from Various Sources**

Data from several government agencies, including welfare, health, and census departments, must be gathered and combined by the system. This guarantees that all pertinent data regarding vulnerable groups is accessible in a single spot, connected for insightful analysis.

- **AI-Powered Classification and Prediction Algorithms**

Based on demographic and health data, the platform should employ AI techniques like machine learning to determine which groups are at risk. In order to assist officials in effectively planning and allocating resources, it must also forecast future welfare demands.

- **GIS Visualizations in Real Time**

The location and density of susceptible people must be shown on real-time, interactive maps and heatmaps that are provided by the system. Users can know where assistance is most required at any given time thanks to this.

- **Policymakers' Interactive Dashboards**

Key insights should be visible in straightforward dashboards for welfare professionals and policymakers. Making educated judgments will be made simpler with the use of these dashboards, which will enable filtering, trend viewing across time, and alert access.

- **Data Export and Report Production**

Data exporting and the creation of thorough reports that may be distributed to interested parties or utilized for additional research, oversight, and auditing should be supported by the system.

- **Strong User Authentication and Security**

The platform needs to include robust security protocols. By using role-based access restrictions and secure login procedures, it should guarantee that only authorized individuals have access to sensitive data.

13.2 Non-Functional Requirements

- **Scalability**

The system must be designed to handle increasing amounts of data and integrate new data sources easily without affecting performance.

- **High System Availability and Performance**

The platform should be reliable and available for use when needed, with fast processing and response times to support real-time decision-making.

- **User-Friendly Interfaces**

Since many users may not be technical experts, the system's interface should be intuitive and simple to navigate, requiring minimal training.

- **Compliance with Data Privacy Laws**

The platform must adhere strictly to Sri Lanka's data protection regulations, ensuring user and beneficiary data is handled responsibly and confidentially.

- **Reliability in Data Processing and Visualization**

The system should consistently produce accurate analyses and visualizations, ensuring the data presented is trustworthy and up-to-date.

14. Commercialization

14.1 Budget

An anticipated cost breakdown for creating and sustaining the suggested AI-GIS Integrated Platform is shown in the budget table that follows. The costs are estimates that could change based on service providers, system size, and vendor price.

Table 2: Budget Breakdown

Description	Cost	Occurring
Server Hosting (Cloud VPS with GIS support)	LKR 15,000	Monthly
App Hosting (iOS Developer Program)	LKR 7,750	Annual
Play Store Hosting (Google Play Developer)	LKR 30,750	One-Time
Cloud Database & GIS Data Storage (AWS S3 / Firebase / ArcGIS Online)	USD 0.03 per GB	Monthly
Domain Registration & SSL Certificate	LKR 5,000	Annual
Third-Party API Costs (GIS maps, Health Data, AI Models)	LKR 20,000	Monthly
AI & GIS Software Licenses (TensorFlow, ArcGIS APIs, QGIS plugins, etc.)	LKR 40,000	Initial Setup
Marketing & Awareness Campaigns	LKR 50,000	Initial Setup
Maintenance & Technical Support	LKR 30,000	Monthly

14.2 Commercialization Methods

Both government agencies and non-governmental organizations (NGOs) can use the proposed AI-GIS Integrated Welfare Platform, which would be presented as a scalable digital governance solution. Data-driven service models, B2G alliances, and NGO partnerships are the main pillars of the commercialization strategy:

1. B2G (Business-to-Government) Partnerships:

Work together with the Ministry of Social Services, provincial councils, and divisional secretariats to incorporate the system into their welfare management procedures.

Offer government agencies subscription-based licensing or service agreements with tiers of pricing according on feature access and demographic coverage.

2. NGO and Development Partner Collaborations:

Collaborate with international development groups, NGOs, and UN agencies to support aid distribution and welfare targeting.

Provide project-based deployments for programs aimed at reducing poverty, disaster relief, and humanitarian help.

3. Direct Service for Policymakers and Researchers:

Offer an analytics dashboard with predicted insights driven by AI and vulnerability mapping based on GIS.

Provide several subscription packages for academic institutions, academics, and policy think tanks that require access to prediction tools and anonymized datasets.

4. Market Expansion:

With an emphasis on high-need areas (urban poor, rural communities, and disaster-prone regions), the initial deployment was made in Sri Lanka.

gradual expansion into South Asian nations with same issues with welfare distribution.

5. Revenue Streams:

subscription or service charges from NGOs and government agencies.

granting third-party digital platforms licenses for AI + GIS prediction models.

specialized reporting and analytics services for international organizations and policymakers.

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